

AUTOMATED FILE ANALYSIS TOOL FOR WINDOWS DIGITAL FORENSICS

Detection of Anti-Forensic Techniques and Hidden Data Artifacts

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Problem Statement

Digital forensic investigators face increasing challenges with sophisticated anti-forensic techniques including file extension spoofing, steganography, and timestamp manipulation. **Manual analysis is time-consuming and prone to human error.** Current tools lack integrated detection capabilities for multiple anti-forensic methods. There is a critical need for automated tools that can rapidly identify file type mismatches, hidden data, and metadata anomalies across large datasets while maintaining forensic integrity and providing clear evidence documentation.

Key Objectives

- Implement file signature analysis module to detect extension-content mismatches using magic numbers
- Develop Shannon entropy calculator to identify encrypted/compressed data and steganography indicators
- Create EXIF metadata parser to detect timestamp anomalies and manipulation evidence
- Build cryptographic hash verification system (MD5, SHA-256) for file integrity checking
- Design interactive visualization dashboard showing entropy distributions and timeline correlations
- Conduct accuracy evaluation using CFReDS and Digital Corpora datasets with ROC analysis
- Perform usability testing with forensic practitioners measuring task completion time and accuracy

Key References

- Beebe, Nicole Lang, and Jan Guynes Clark. “Digital Forensic Text String Searching: Improving Information Retrieval Effectiveness by Thematically Clustering Search Results.” Digital Investigation, vol. 4, Sept. 2007, pp. 49–54, <https://doi.org/10.1016/j.diin.2007.06.005>.
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- Chen, Fuhai, et al. StructCap. 19 Oct. 2017, <https://doi.org/10.1145/3123266.3123275>. Accessed 29 Aug. 2023.
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Introduction

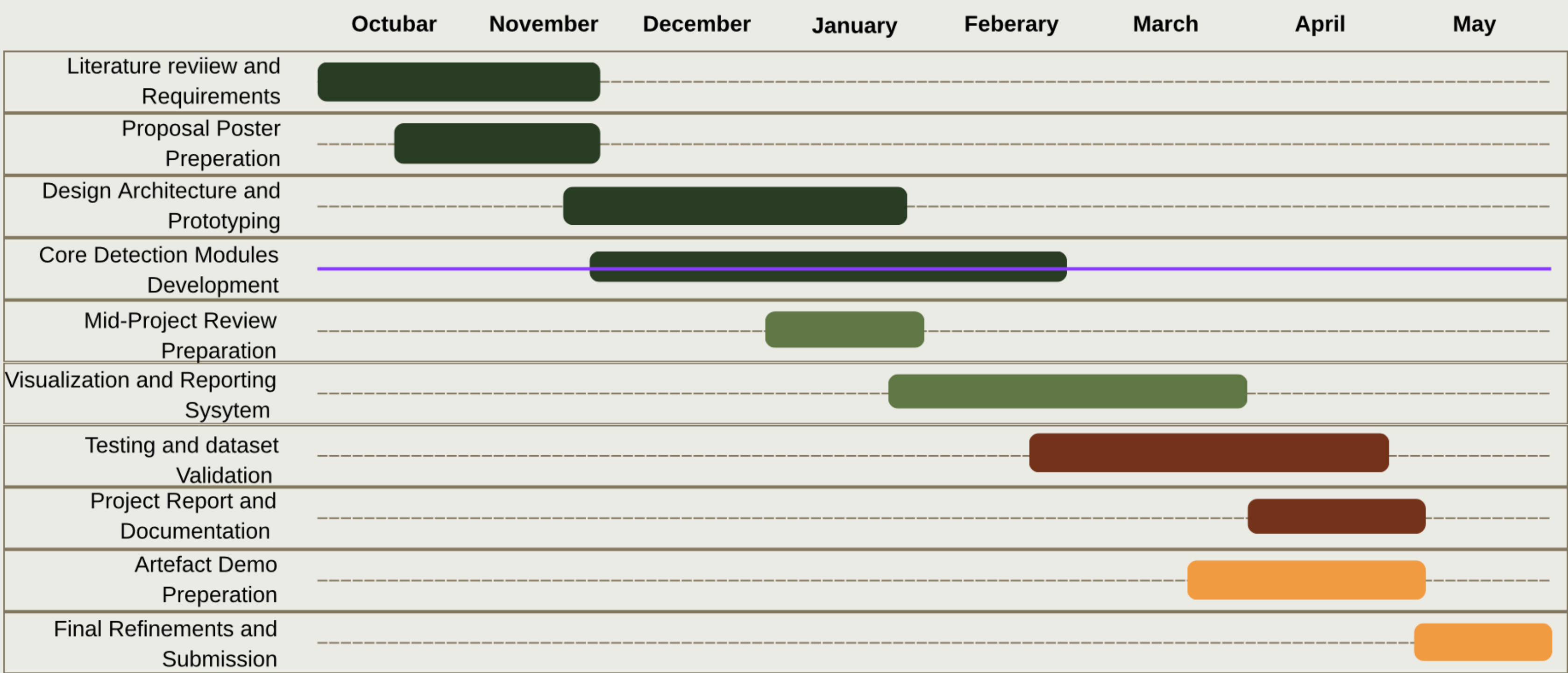
Cybercriminals increasingly employ anti-forensic techniques to conceal evidence, with 67% of digital investigations encountering file obfuscation methods. This project addresses the critical gap in automated detection tools by developing a **Windows-based forensic analyzer** that combines multiple detection algorithms. The tool targets law enforcement agencies, corporate security teams, and forensic laboratories requiring rapid triage capabilities. By integrating **signature analysis**, **entropy detection**, and **metadata examination** into a single automated workflow, the system will reduce analysis time by an estimated 70% while improving detection accuracy. The solution maintains forensic integrity through **hash verification** and comprehensive audit logging, making findings court-admissible.

Timeline Overview

- October - November 2025: Foundation phase with literature review, requirements and gathering, and proposal poster preparation
- 17 November 2025: Project proposal Poster
- November - January 2026: Design and architecture development with prototyping
- December - February 2026: Core detection modules implementation (signature analysis, entropy calculation, metadata parsing)
- 12-26 January 2026: Mid-project review with supervisor
- February - March 2026: Visualization and reporting system development
- March - April 2026: Testing with forensic datasets and validation
- April - May 2026: Documentation, artefact demonstration preparation, and final submission

PROJECT PLAN OVERVIEW

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● Initiation & Planning ● Development ● Execution & Monitoring ● Closing

Project Aims

- Design an automated file analysis tool for Windows that detects anti-forensic techniques
- Implement multi-method detection combining signature analysis, entropy examination, and metadata verification
- Develop comprehensive reporting capabilities with visualizations for forensic investigators

Legal, Ethical and Professional Considerations

- All analysis operations will be logged with cryptographic hashes to maintain chain-of-custody for court admissibility.
- Testing will use publicly available forensic datasets (CFReDS, Digital Corpora) and synthetic data only – no real case data.
- The tool will include clear usage warnings about legal authorization requirements before analyzing systems.
- Development adheres to the BCS Code of Conduct, emphasizing public interest and professional competence.
- Privacy safeguards ensure no personal data extraction beyond forensic necessity.
- The system is designed strictly for authorized forensic use only.

Use of AI

- AI tools (GitHub Copilot, ChatGPT) will assist with code generation for standard algorithms and documentation drafting.
- All AI-generated code will be manually reviewed, tested against known datasets, and validated for accuracy before integration
- I'll use AI as a debugging partner as some of the tools are really helpful for debugging.
- I also use AI during brainstorming to explore different forensic approaches, but I still choose the ideas that make sense for my project and justify why they fit the legal and technical requirements.